



Department of Information Technology

Academic Year 2025– 2026 (Odd Semester)

Degree, Semester & Branch: V Semester B.Tech. IT

Course Code & Title: CCS334 & Big Data Analytics

Name of the Faculty member (s): Mrs.M.Rethina Kumari

Innovative Practice Description

Unit / Topic: Unit 3/Job Scheduling

Course Outcome: CO3

Activity Chosen: Case Based Learning

Justification:

Case-Based Learning (CBL) enables students to analyze real-world job scheduling problems faced in Big Data environments like Hadoop and Spark.

The activity develops students' analytical, problem-solving, and critical thinking skills by comparing different scheduling algorithms such as FIFO, Priority, and Fair Scheduling.

It encourages innovative thinking as students propose their own hybrid scheduling ideas to improve fairness, efficiency, and performance in distributed systems.

Students connect theoretical knowledge of Big Data frameworks with practical job scheduling issues encountered in real data centers.

Time Allotted for the Activity: 20 minutes

Details of the Implementation:

The instructor provided an overview of **Job Scheduling concepts** in Big Data systems explaining **FIFO and Priority Scheduling** policies.

A **case scenario** was presented to all students: A Big Data company has five jobs with different arrival times, priorities, and memory needs. Which scheduling approach is most efficient and why?"

Students were divided into **teams of 5–6 members**. Each team received the following data:

Job	Arrival (s)	Execution Time (s)	Memory (GB)	Priority
J1	0	10	4	High
J2	1	20	8	Medium
J3	2	15	6	High
J4	3	8	2	Low
J5	4	12	5	Medium

Step 1 – Think (2 minutes): Students individually analyzed the given dataset, recalling the logic of each scheduler (FIFO, Priority).

Step 2 – Pair / Team Work (8 minutes): Teams created simple Gantt charts for **FIFO** and **Priority Scheduling**, calculated **average completion time or makespan**, and discussed which scheduler performed better. They proposed one short improvement idea (e.g., “hybrid scheduler combining priority with resource-awareness”).

Step 3 – Share (5 minutes): Selected teams presented their comparison and improvement ideas to the class. The instructor summarized key findings, related them to **YARN Fair and Capacity Schedulers**, and emphasized trade-offs between fairness and performance.

Step 4 – Reflection (5 minutes): Students shared takeaways: the importance of scheduling fairness, how resources affect job order, and how real-world clusters optimize throughput.

CO – PO / PSO mapping:

(1 – Low 2 – Moderate 3 – High)

CO	PO1	PO2	PO3	PO9	PO10	PSO1
CO 3	2	1	1	1	1	1

PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO1
	2	1	1	1	1	1
Justification for correlation	Strengthens analytical and problem-solving skills by evaluating scheduling algorithms.	Encourages application of theoretical concepts to real-world data processing scenarios.	Fosters innovative thinking through the design of improved or hybrid scheduling strategies	Promotes teamwork and collaborative analysis during group discussions	Improves communication and presentation skills through sharing findings and solutions	Aligns with practical implementation of Big Data tools like Hadoop YARN and Spark schedulers

Images / Screenshot of the practice:



Figure1 and 2 shows discussion of Students Gowtham -Gopi & Abinaya.G.M-Ashika pair explaining their discussion to the class.

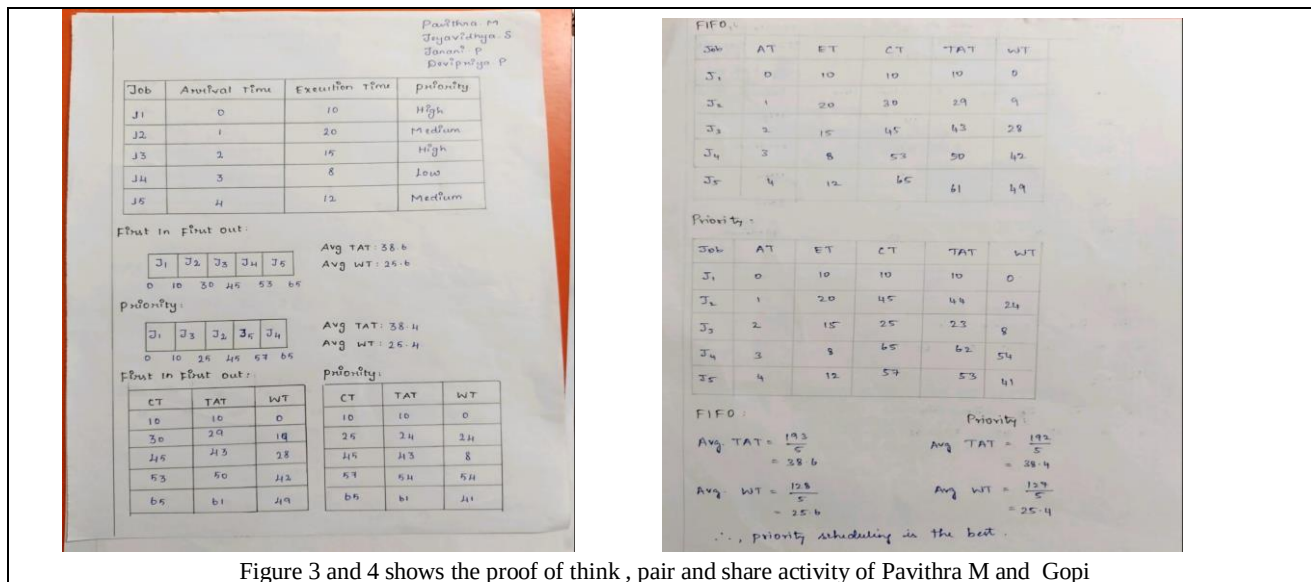


Figure 3 and 4 shows the proof of think, pair and share activity of Pavithra M and Gopi

Glimpse of Think Pair Share Activity

❖ Feedback of practice from students and other stakeholders:

- Students found the session interactive and relevant to industry applications.
- The Think-Pair-Share format helped in **breaking down complex business strategies** linked to mobile BI.
- Peer discussions clarified doubts on technical integrations (e.g., GPS tracking, cloud sync).

❖ Benefit of the practice:

- Students gained **confidence in explaining mobile BI features**.
- Improved **classroom engagement and participation**.
- Strengthened understanding of **how intelligent apps assist real-time decision-making**.

❖ Challenges faced in implementation:

- A few students found it difficult to relate technical tools with business cases.
- Some teams needed additional prompting to initiate discussions.
- Time was slightly insufficient for in-depth idea sharing—extra 5 minutes would improve effectiveness.

References:

1. <https://www.readingrockets.org/classroom/classroom-strategies/think-pair-share>
2. <https://www.wgu.edu/blog/how-think-pair-share-activity-can-improve-your-classroom-discussions1712.html>
3. <https://www.prodigygame.com/main-en/blog/think-pair-share-learning-strategy>

Signature of Faculty Member

HOD